Do oil shocks drive the business cycle?

An Internal-Instrument Structural VAR Approach

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1 Introduction

The recent inflation surge has brought renewed attention to a classic question in macroeconomics: how much do energy price shocks feed into consumer prices and wages? Bernanke and Blanchard (2024, henceforth BB) explain the U.S. pandemic-era inflation with a dynamic model of prices, wages, and inflation expectations, both in the short-run and long-run. Their main conclusion is that most of the inflation surge came from shocks to prices given wages, especially commodity and energy prices and pandemic shortages, rather than from an overheated labour market. Their identification relies on treating energy prices, food prices, the shortage index, and the vacancy-to-unemployment ratio as exogenous regressors, and assuming wages react to the other endogenous variables only with a one-quarter lag. This project revisits that analysis inside a unified structural vector autoregression (SVAR). Instead of assuming that energy prices are exogenous, they are brought inside the system, and the oil shock is identified with an external instrument. Assuming energy prices and the shortage index are exogenous is a strong requirement; it means these variables must be uncorrelated with the structural shocks driving wages and prices. This assumption is questionable for energy prices, which are plausibly endogenous and shaped by both demand and supply forces. Moreover, the single-equation approach used by BB fails to fully capture the simultaneous interactions among the endogenous variables. The oil shock is identified using the oil supply news series of Känzig (2021), built from high-frequency movements of oil futures prices in a tight window around OPEC announcements. This paper examines whether BB's conclusions about the oil channel survive when energy prices are endogenous and identified using independent, external variation.

A few clear patterns emerge from the analysis. A positive oil supply news shock raises the relative price of energy by about 8.7 percentage points on impact and lifts CPI inflation by roughly 0.84 pp on impact and 0.62 pp after one quarter, but the effect is short-lived, as it is essentially gone within four to five quarters. The same shock leaves wage growth statistically indistinguishable from zero at every horizon. The forecast-error variance decomposition (FEVD) tells the same story, attributing approximately 29% of the two-year variance of CPI inflation to the oil supply shock, but only 0.6% of the variance of wage growth. These results reproduce, within a fully simultaneous SVAR, the two pillars of BB's account: energy shocks are a powerful and fast-acting source of headline-price movements, while their impact on wages and inflation expectations is limited.

2 Data

All series are quarterly and span the pre-COVID window 1990Q1-2019Q4, following BB (2024) who document a break in inflation dynamics around 1990, so that the estimation reflects the same pre-pandemic situation.

Table 1: Data dictionary.

Variable	Mnemonic	Description	Source / ID
Oil supply news	oilnews	Monthly OPEC-announcement future surprises, summed within quarter	Känzig (2021)
Energy rel. price	Δrpe	$400 \cdot \Delta \log$(CPI energy/ECI)	BLS CPIENGSL ÷ ECI
CPI inflation	π	$400 \cdot \Delta \log$(CPI), quarterly avg.	BLS CPIAUCSL
Wage growth	Δw	ECI 3-month % change × 4 (annualised)	BLS CIU2010000000000Q
1-yr expectations	π^e_1	One-year expected inflation	Cleveland Fed
10-yr expectations	π^e_{10}	Ten-year expected inflation	Cleveland Fed

The VAR system collects six endogenous variables, ordered as:

$$y_t = [\text{oilnews}_t,\ \Delta\text{rpe}_t,\ \pi_t,\ \Delta w_t,\ \pi^e_{1,t},\ \pi^e_{10,t}]'$$

CPI inflation (π) is the quarterly annualised change in the BLS CPI (CPIAUCSL). Wage growth (Δw) is the quarterly annualised change in the Employment Cost Index (ECI). The ECI is generally thought to be a better measure of wages than average hourly earnings because it corrects for changes in average earnings due to changes in the composition of employment (Bernanke and Blanchard, 2024). Relative energy-price inflation (Δrpe) is built as $400 \cdot \Delta \log$(CPI-energy/ECI level), which stands for the annualised log-change in the price of energy measured relative to wages. Inflation expectations (π^e_1, π^e_{10}) represent the one-year and ten-year series from the Federal Reserve Bank of Cleveland, converted to percent and averaged to quarterly frequency.

Unlike BB, the relative price of food is excluded to keep the system parsimonious at n = 6 variables; the analysis focuses on the energy channel, which BB identify as the dominant price shock.

Table 2: Descriptive statistics, estimation sample 1990Q1–2019Q4 ($T = 120$).

Variable	**Mean**	**Std**	**Min**	**Median**	**Max**
oilnews	0.006	0.970	-2.955	-0.062	2.696
Δrpe (ann. %)	-0.001	18.904	-106.597	1.916	50.781
π CPI inflation (ann. %)	2.391	1.945	-9.267	2.571	6.850
Δw wage growth (ann. %)	2.823	1.084	0.400	2.800	5.200
π^e_1 1-yr exp. (%)	2.399	0.820	0.125	2.335	4.950
π^e_{10} 10-yr exp. (%)	2.504	0.698	1.475	2.406	4.016

The Känzig instrument spans 1975M01–2020M12 and, summed to quarters, has mean almost equal to 0 and unit standard deviation. Summing preserves the total news content within the quarter, since the shocks are surprises (Känzig, 2021).

The binding sample constraint is CPI energy. After aligning all series and initialising four lags, the VAR(4) is estimated over 1990Q1–2019Q4, a sample of 120 quarterly observations. Summary statistics confirm the expected scales: CPI inflation averages 2.39% (s.d. 1.95, min −9.27 in the 2008 collapse), wage growth 2.82% (s.d. 1.08), and one-year and ten-year expectations 2.4% and 2.5%; Δrpe is an order of magnitude more volatile (s.d. 18.9) than every other series, reflecting the well-known volatility of energy prices. Bernanke and Blanchard (2024) interpret commodity price movements as driven primarily by global demand factors, suggesting that treating energy prices as exogenous regressors may be unsatisfactory.

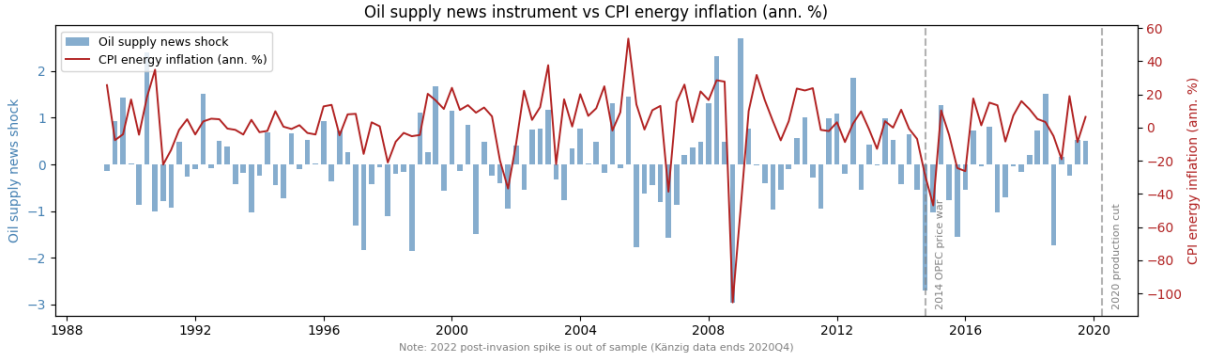


Figure 1: *Quarterly oil supply news instrument (bars, left axis) and CPI energy inflation (line, right axis). Dashed lines mark the 2014 OPEC price war and the 2020 OPEC+ production cut.*

Figure 1 displays the aggregated oil supply news instrument (bars) against the relative CPI energy inflation (red line). The two track each other closely around key historical episodes: positive news (production-cut surprises) line up with energy price spikes, while during the 2014 OPEC price war, when Saudi Arabia chose to defend market share rather than cut production, the oil news shock turned sharply negative, correctly anticipating the collapse in energy prices that followed (CPI energy inflation drops to about -40% in 2015). The same pattern appears in early 2009, when OPEC cut output by more than 4 million barrels per day to support prices during the financial crisis.

This visual co-variation is the first evidence that the instrument is relevant for the variable it is meant to identify.

3 Model and Identification

A reduced-form VAR(p) is estimated on the six endogenous variables described in Section 2:

$$y_t = c + A_1 y_{t-1} + \ldots + A_p y_{t-p} + u_t, \qquad u_t \sim \mathcal{N}(0, \Sigma).$$

The reduced-form residuals are linear combinations of n orthogonal structural shocks, $u_t = B_0^{-1}\varepsilon_t$, and structural shocks are recovered with a Cholesky factorisation of Σ, $\varepsilon_t = P^{-1}u_t$ with P the lower-triangular Cholesky factor ($PP' = \Sigma$).

3.1 Lag selection and stability

The number of lags is selected on the basis of information criteria and, foremost, residual adequacy. The information criteria (AIC, BIC, FPE, HQIC) are minimised at p = 1,

reflecting the parsimony penalty in a system with n = 6 variables. At p = 1 the residuals remain serially correlated, and since the primary object of interest is the set of impulse responses, a lag order that leaves autocorrelation in the residuals would distort the dynamics the model is meant to recover. The choice therefore weights residual adequacy more heavily than the information criteria.

Residual adequacy is checked using two tests: the multivariate Portmanteau test, which jointly tests whether all residual autocorrelations up to lag h are zero across the entire system, and the univariate Ljung-Box Q-test, which tests equation by equation whether residuals are serially uncorrelated up to lag h. Both tests have null hypothesis of no autocorrelation; rejection means the chosen lag order leaves unexplained dynamics in the residuals. At $p = 1$, the Ljung-Box test rejects in three of six equations, indicating that the model underfits the dynamics. At p=4, only the oilnews equation rejects, which is attributable to the sparse OPEC-event-driven structure of the instrument rather than to any flaw in the specification of the macroeconomic equations. All economically relevant equations (CPI inflation, wage growth, expectations) pass at $p = 4$.

The multivariate Portmanteau test rejects at all lag orders, consistent with structural instability over a 30-year sample rather than a pure lag-length problem.

Table 3: Lag-order diagnostics.

Lag p	AIC	BIC	FPE	HQIC	Portmanteau p	LB rejections (of 6)
1	-3.100^*	-2.119^*	0.04510	-2.701^*	0.0010	3: *oilnews*, Δw, π^e_{10}
2	-3.060	-1.229	0.04713	-2.317	0.0025	2: Δw, π^e_{10}
3	-2.887	-0.195	0.05673	-1.794	0.0005	1: π^e_{10}
4	-2.848	0.712	0.06037	-1.403	0.0004	1: *oilnews*
5	-2.919	1.521	0.05859	-1.117	0.0023	0: —
6	-2.655	2.674	0.08137	-0.492	0.0002	0: —
7	-2.421	3.806	0.11315	0.106	0.0000	0: —
8	-2.510	4.626	0.11877	0.385	0.0000	0: —

The choice lands on p = 4 for multiple reasons. (i) Adequacy: at p=4 the only remaining rejection is the instrument equation, while the wage-growth and long-run expectations equations that reject at shorter lags are clean. (ii) Parsimony: p=5 is the first lag order where no equation rejects the Ljung–Box test, but going beyond four lags adds more parameters and decreases degrees of freedom, reducing the precision of the impulse responses. (iii) Comparability: for quarterly lags the natural choice according to the rule of thumb is p=4, which is the same as Bernanke and Blanchard's lag structure, making the model directly comparable to theirs.

Augmented Dickey-Fuller tests reject a unit root for oilnews, Δrpe and π, while wage growth and the two expectations series show evidence of persistence. Despite this, the VAR is estimated in levels rather than in differences. This follows the established tradition of Sims (1980), who argued that differencing throws away long-run information, and the practical guidance of Kilian and Lütkepohl (2017), who show that VAR estimates in levels remain consistent even in the presence of unit roots or cointegration, while differencing introduces misspecification if the variables share long-run relationships. Estimating in levels is also in line with BB's approach, who apply the same specification over the same sample period.

The dynamic adequacy of the VAR model is assessed by examining the characteristic roots of the system.

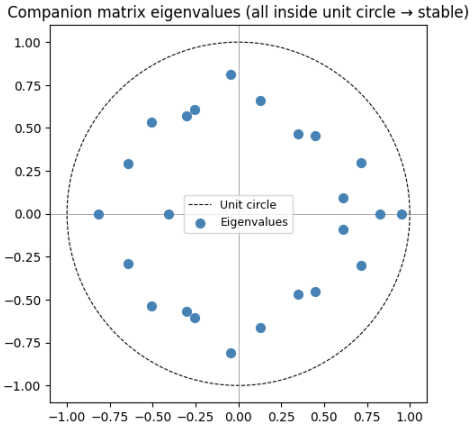


Figure 2: *Companion matrix eigenvalues. All eigenvalues lie inside the unit circle, confirming VAR(4) stability.*

Figure 2 plots the companion-matrix eigenvalues against the unit circle. The fitted VAR(4) is dynamically stable, as all companion eigenvalues lie strictly inside the unit circle, the largest having modulus 0.95. Equivalently, all characteristic roots of the system lie outside the unit circle, the smallest having modulus 1.05, leaving a sufficient margin from the boundary.

3.2 Identification strategy

Identification builds on the logic of external instruments developed by Stock and Watson (2012) and Mertens and Ravn (2013), using the Känzig (2021) oil supply news series. Rather than running a separate proxy-SVAR, the instrument is embedded inside the VAR and ordered first under a Cholesky structure. The reduced-form residuals u_t are related to the structural shocks ε_t by $u_t = B_0^{-1}\varepsilon_t$, where B_0^{-1} is the contemporaneous impact matrix. As introduced at the beginning of Section 3, to recover the structural shocks, a Cholesky decomposition is applied to the estimated residual covariance matrix $\hat{\Sigma}$: writing $\hat{\Sigma} = PP'$ where P is lower triangular, the structural shocks are recovered as $\varepsilon_t = P^{-1}u_t$.

The key identifying assumption is the ordering of the variables. The Känzig (2021) oil supply news instrument is placed first in the vector y_t. Under the Cholesky ordering, this means no other structural shock moves *oilnews* contemporaneously, while the oil supply news shock is free to affect all other variables on impact.

Since Känzig surprises are constructed from high-frequency futures price movements in a tight window around OPEC announcements, they are orthogonal to current U.S. macroeconomic conditions by construction, making the first Cholesky shock a valid measure of exogenous oil supply news. As Stock and Watson (2012) note, with the instrument ordered first the impulse responses to that shock are invariant to the ordering of the remaining variables.

Following BB (2024), who imposed restrictions on the contemporaneous relationships among the variables based on their simple model, the ordering of the remaining variables adopts the same timing assumptions: Δrpe_t is ordered second, as global energy prices are driven by world markets and unlikely to respond contemporaneously to U.S. wages or prices; π_t comes before Δw_t, reflecting BB's assumption that prices can affect wages contemporaneously while wages respond to prices only with a lag; inflation expectations

are ordered last, as they react to observed prices and wages rather than driving them within the quarter.

This approach relaxes the key assumption in BB (2024), who treat energy prices as an exogenous regressor. Here, Δrpe_t is endogenous and inside the system; the oil shock is still cleanly identified through externally validated variation in OPEC announcement surprises. As BB (2024) note, the identification of the wage equation is achieved by assuming that wages respond to other variables with a lag of one quarter. Wage inflation then affects inflation, and by implication, inflation expectations contemporaneously. The internal-instrument approach here replaces that equation-by-equation timing restriction with a system-wide recursive structure.

4 Results

4.1 Impulse responses to oil supply news shocks

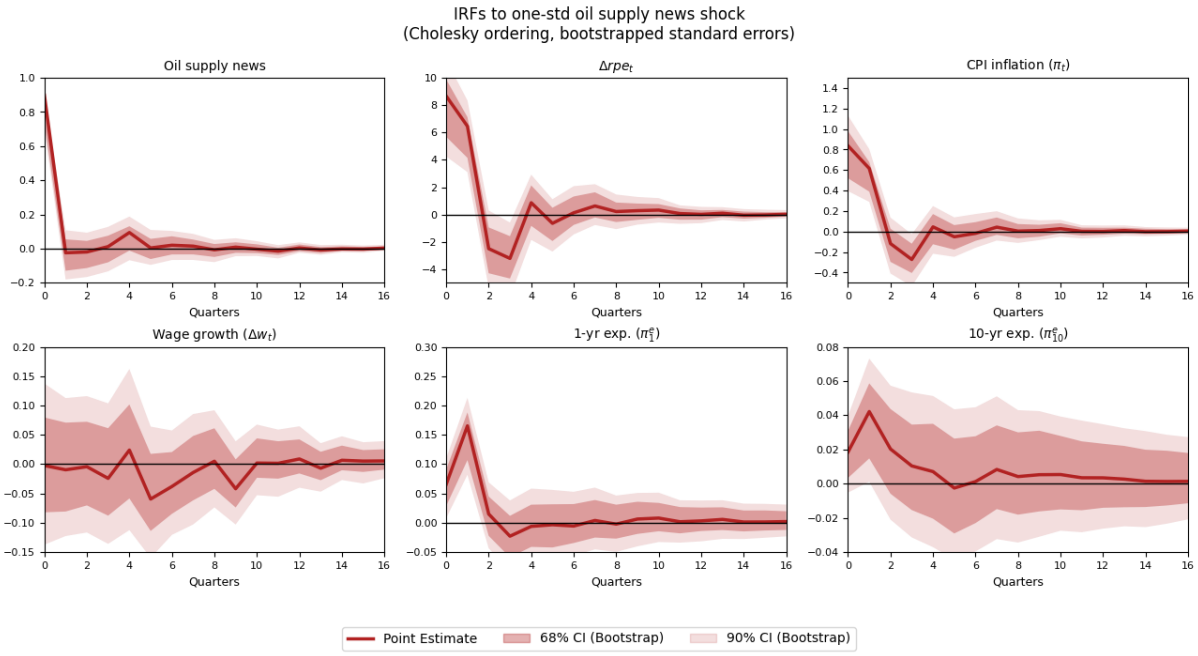


Figure 3: *Impulse response functions to a one-standard-deviation oil supply news shock (Cholesky ordering). Shaded bands show 68% and 90% bootstrap confidence intervals (recursive bootstrap, 500 replications)*

Figure 3 displays the dynamic effects of a one-standard-deviation oil-supply news shock. The relative price of energy jumps to 8.7 pp on impact and reverts after one quarter. CPI inflation rises by 0.84 pp on impact and 0.62 pp at h=1, then returns to zero after the fourth quarter. The response is large but short-lived. Inflation expectations rise on announcement (one-year expectations peak at 0.17 pp at h=1, ten-year expectations peak at 0.04 pp) and both decay quickly toward zero. Bernanke and Blanchard (2024) find a very similar pattern: the effects of an energy price shock on inflation largely disappear within one quarter, leaving only a small and persistent increase of around 0.1 pp due to imperfectly anchored expectations.

Wage growth shows no meaningful response at any horizon; Bernanke and Blanchard (2024) show that real wage rigidity (the catch-up effect) is limited and that short-run and long-run inflation expectations appear to be reasonably well anchored, adjusting very slowly in response to actual inflation. The low persistence of inflation reflects the weak catch-up effect: workers are unable to recover the effects of unexpected inflation on their real wages. This is consistent with the near-zero wage response found here: without a catch-up mechanism or de-anchored expectations, energy shocks do not generate a wage-price spiral.

The combination of a sharp increase in energy prices and consumer prices, rising inflation expectations, and an attenuated real-wage reaction matches the pattern found by Känzig (2021). He finds that oil supply news shocks cause consumer prices and inflation expectations to rise, providing evidence for a strong channel operating through supply expectations, and that these effects occur even if current oil production does not move. Bernanke and Blanchard (2024) highlight that this stands in sharp contrast to the 1970s, when less anchored expectations and stronger catch-up mechanisms made oil price shocks far more persistent.

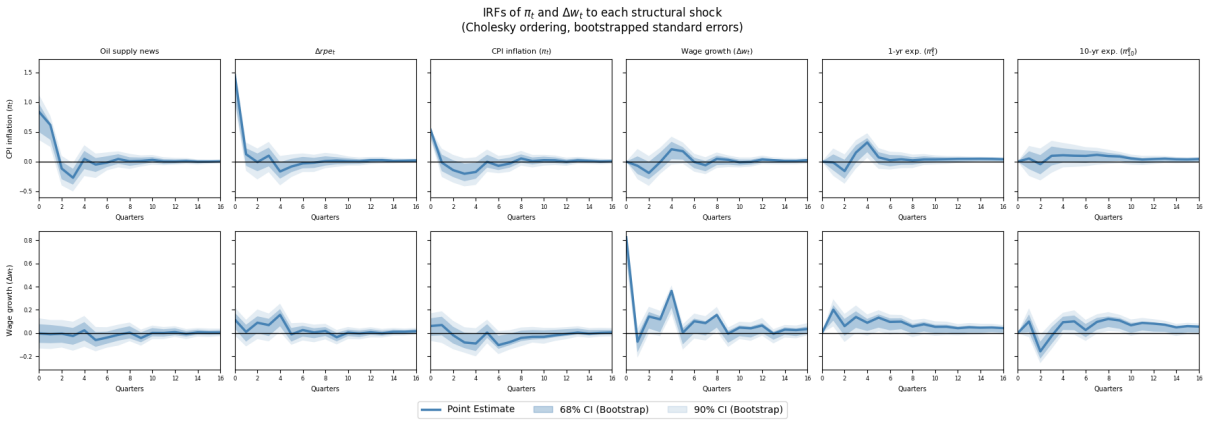


Figure 4: *Impulse response functions of π_t and Δw_t to each structural shock (Cholesky ordering). Shaded bands show 68% and 90% bootstrap confidence intervals (recursive bootstrap, 500 replications).*

Figure 4 shows the responses of CPI inflation and wage growth to all six shocks. Inflation reacts substantially to oil supply news, the energy relative price shock, and of course to its own shock. Wage growth, instead, responds significantly only to its own shock, with weak deviations in the others, and showing almost no reaction to oil supply news. The oil channel heavily affects prices but barely influences wages, consistent with Bernanke and Blanchard (2024), who find that energy price shocks are among the largest contributors to pandemic-era price inflation, while their contribution to wage growth is substantially smaller.

4.2 Forecast error variance decomposition (FEVD)

Table 4 reports the share of each variable's forecast-error variance attributable to the oil supply news shock across horizons from one quarter to two years (8 quarters).

The shock explains a substantial share of energy price and CPI inflation variance. The two series start close together at one quarter, at 22.6% for CPI inflation and 24.7% for

the relative energy price, then rise to about 31% and 33% respectively by the one-year horizon, supporting BB's finding that energy prices are a primary driver of pandemic-era price inflation. Bernanke and Blanchard (2024) estimate the long-run effect of an energy price shock on the price level at around 10%, substantially larger than energy's 7.4% share in the CPI basket. While their estimate concerns the price level and the FEVD here addresses variance, both identify energy as a primary driver of inflation. Wage growth, by contrast, is barely affected: the shock explains almost none of its variance (negligible at one quarter, only 0.6% after two years), reflecting their conclusion that wage-price spirals are not the main transmission channel. This near-zero share mirrors BB's own wage equation, where the estimated catch-up coefficient is small and statistically insignificant (-0.024 with $p = 0.76$), the structural reason an energy shock does not propagate into wages.

Inflation expectations are intermediate, peaking at around 19% for 1-year expectations at the two-quarter horizon before settling to 16% at one year and about 10% by two years, while 10-year expectations remain low throughout all horizons (peaking only at 3.4%), indicating well-anchored long-run expectations.

Table 4: FEVD – Share explained by oil supply news shock (%)

Variable	$h = 1q$	$h = 2q$	$h = 3q$	$h = 4q$	$h = 5q$	$h = 6q$	$h = 7q$	$h = 8q$
oilnews	100.0	82.2	79.7	78.3	73.8	73.3	72.9	72.7
Δrpe (rel. energy price)	24.7	33.4	32.6	33.2	31.9	31.4	31.2	31.1
π (CPI inflation)	22.6	30.8	30.3	31.0	29.4	29.0	28.9	28.8
Δw (wage growth)	0.0	0.0	0.0	0.1	0.1	0.5	0.6	0.6
π^e_1 (1-yr expectations)	5.3	19.3	18.1	15.7	13.4	11.9	11.0	10.4
π^e_{10} (10-yr expectations)	1.4	3.4	3.0	2.5	2.1	1.8	1.6	1.5

Figure 5 shows the full variance decomposition at the 8-quarter horizon. Each panel shows the share of a variable's forecast-error variance attributable to each of the six structural shocks, and shares within a panel sum to 100%.

Oil supply news drives energy prices and CPI inflation considerably, but wage growth is almost entirely driven by its own shock. The oil-wage transmission is essentially absent. The 10-year expectations are dominated by their own dynamics, in accordance with the view that central bank credibility kept long-run expectations anchored despite the energy shock.

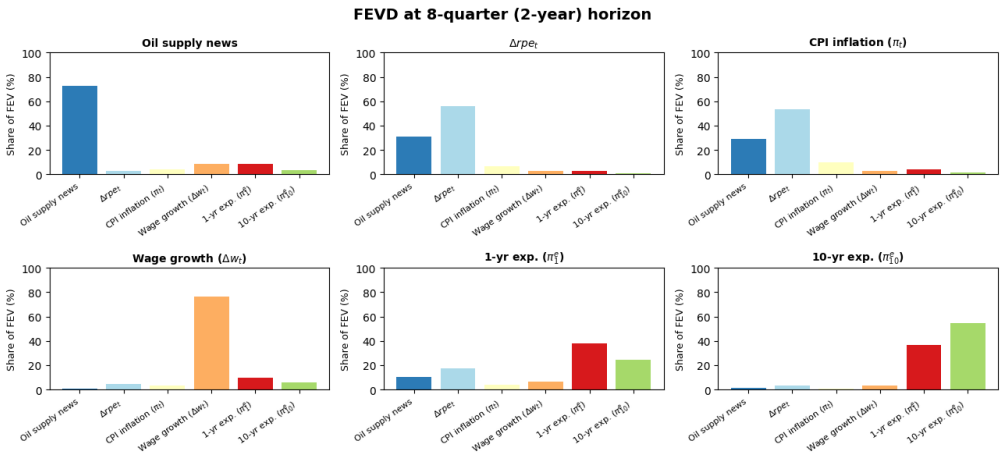


Figure 5: FEVD at 8-quarter (2-year) horizon

4.3 Historical decomposition

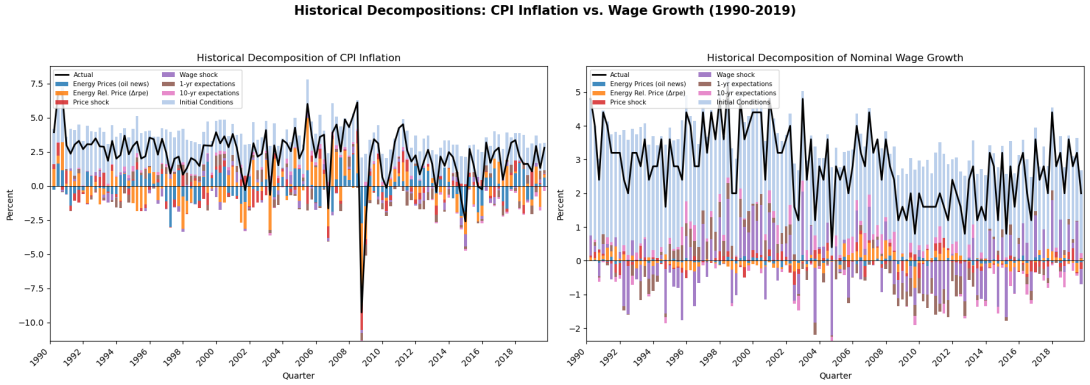


Figure 6: *Historical Decomposition of CPI inflation and wage growth — Cholesky VAR(4), 1990Q1–2019Q4*

The historical decomposition, described in Lütkepohl (2005), breaks down the actual path of inflation and wages into the contribution of each structural shock quarter by quarter. The bars sum exactly to the black line.

For CPI inflation, oil news and energy relative price shocks (the blue and orange bars) account for most of the high-frequency swings throughout the sample. This is especially visible around 2009, where the sharp deflation and the following rebound are almost entirely driven by energy. This is congruent with what BB document: energy shocks are the principal mover of overall inflation around major turning points.

The wage growth panel displays a completely different story. The oil-related contributions are barely visible, since wages are governed by their own shock, short-run expectations, and a large stable initial-conditions component sitting around 3% throughout the sample, with some peaks at 5%. That near-constant trend, at about 3%, closely aligns with the wage norm Bernanke and Blanchard identify in their analysis.

So the historical decomposition reproduces their central asymmetry: energy shocks explain a lot of inflation variation but almost none of the wage variation, which is why the post-2020 inflation surge did not trigger a wage-price spiral.

5 Conclusion

Identifying oil supply news shocks with the Känzig (2021) instrument inside a six-variable Cholesky SVAR, this paper finds that such shocks have large but short-lived effects on U.S. inflation and relative energy prices, a smaller and transitory effect on short-run inflation expectations, and essentially no effect on wage growth or long-run expectations. A one-standard-deviation shock raises CPI inflation by about 0.84 pp on impact, with the effect fading within roughly a year. The FEVD attributes about 29% of the two-year CPI-inflation variance to the oil shock, against under 1% of wage-growth variance. The historical decomposition reinforces this picture: energy shocks account for most of the high-frequency swings in CPI inflation, while wage growth is governed almost entirely by its own dynamics and a stable initial-conditions component near 3%. These results confirm, from a different identification strategy, the two core findings of Bernanke and Blanchard (2024): inflation was driven mainly by shocks to prices given wages, and transmission into wages was limited by anchored expectations and a weak catch-up mechanism.

The answer to the title question is nuanced: oil supply news is an important driver of inflation, but not of the wage dynamics that sustain it. Importantly, this conclusion holds even when energy prices are treated as endogenous rather than exogenous as in BB; the identification strategy changes, but the results do not.

Three limitations stand out. First, the supply-chain variable is excluded: since BB find that sectoral shortages were a major driver of pandemic-era inflation, omitting it means the model cannot cleanly separate the energy channel from the shortage channel. Second, identification is recursive: while ordering the instrument first delivers a clean oil shock that is invariant to the ordering, the cross-effects among the remaining variables still depend on the Cholesky ordering, which rests on the strong assumption that within each quarter causality runs one way only, so each variable reacts to those ordered before it but not after. Third, the sample ends in 2019Q4, so the model cannot speak directly to the post-2020 inflation surge that motivates the analysis.

The natural first extension would be to implement a genuine proxy-SVAR with an explicit first-stage F-statistic, which would sharpen inference on instrument relevance beyond what the internal-instrument approach provides. Re-estimating with local projections following Jordà (2005) would further test sensitivity to the dynamic VAR structure, as Känzig (2021) does in his robustness checks; because local projections estimate each horizon with a separate direct regression rather than by iterating the VAR forward, specification errors do not accumulate across horizons, providing a useful robustness check on the longer-horizon responses.

More broadly, the findings suggest that the oil channel operates primarily through prices rather than wages. This is a result that, if confirmed in extended specifications, has direct implications for how central banks should respond to commodity price shocks.

6 References

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